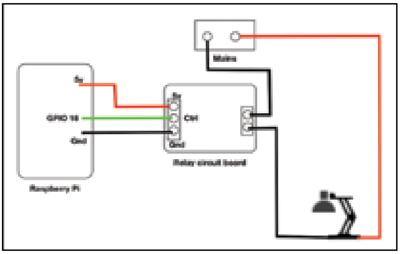
To start with, let's try to control the lights and move further to add more complexity.

Remotely control light

Turning on the lights and turning them back off are things we do throughout the day. Many a time, we forget to switch off electronic appliances when we leave home. It's all good until you forget to switch off your air conditioner or television set, which can effortlessly waste electricity and add to light bill costs. For the purpose of this DIY, we'll look at a simple circuit and code that will help you switch a lamp on and off. This is an extension to the LED blinking program that we saw in the previous chapter.



Basic circuit for switching appliances ON and OFF

Hardware needed:

- Raspberry Pi
- Relay circuit
- Lamp for circuit testing

If you're a beginner, we recommend using a relay circuit from the market. You could also build your own relay circuit using a motor driver or a transistor. In case you want to build your own relay circuit, please take the help of a friend who has adequate knowledge about electronics. Alternatively, you can follow an online tutorial such as the one available at *http://dgit.in/1O98eyK*.

> Be careful when building the relay circuit or connecting the relay circuit. A wrong connection may damage your Raspberry Pi board, harm you or switch off the lights of your house. Please connect with caution and under expert supervision.

First, install the Python GPIO wrapper by using the following command in the terminal:

```
sudo apt-get install rpi.gpio
```

Connect the circuit as shown in the figure. Now, in the terminal enter the following command to create a Python script that will control the appliances:

```
nano basic _ switch.py
```